

1. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        sleep(10);
    } else if (pid > 0) {
        // Parent process
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        wait(NULL); // Wait for child to finish
    } else {
        // Fork failed
        perror("fork failed");
        return 1;
    }

    return 0;
}
```

```
vishnu@VISHNU:~$ gcc practical3a.c -o practical3a
vishnu@VISHNU:~$ ./practical3a
Parent Process
PID : 1060
Child Process
PID : 1061
PPID : 1060
vishnu@VISHNU:~$ nano practical3a.c
vishnu@VISHNU:~$ |
```

- 2.